

FX Hedging as a Contagion Channel

How NBFI Currency Risk Management
Transmits Global Financial Shocks

Project 4 FX Hedging Amplification Channel

Combining Rey et al. (2024) and Nenova, Schrimpf & Shin (2025)

Key References:

Rey, Stavrageva & Tang (2024). Currency centrality in equity markets. NBER WP 33003.

Nenova, Schrimpf & Shin (2025). Global portfolio investments and FX derivatives. BIS WP 1273.

Ando, Greenwood-Nimmo & Shin (2022). Quantile connectedness. Management Science.

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1. Motivation & Research Gap

The global FX derivatives market has grown to \$75 trillion in outstanding notional (BIS OTC statistics, end-2024), driven primarily by non-bank financial intermediaries (NBFIs) -- investment funds, pension funds, and insurance companies -- hedging cross-border bond investments. This growth has created a powerful but underappreciated channel for the international transmission of financial shocks.

Two recent papers illuminate complementary pieces of this mechanism. Rey, Stavrageva & Tang (2024) show that equity market shocks drive exchange rate movements through a 'currency centrality' network, explaining 95% of monthly FX variation vs the USD. Nenova, Schrimpf & Shin (2025) show that FX swap volumes are a barometer of NBFIs risk-taking, with yield curve slopes determining hedging demand and costs.

This project combines both frameworks to identify a complete contagion channel: equity shocks move exchange rates (Rey et al.), which alter FX hedging costs for NBFIs (Nenova et al.), triggering portfolio adjustments that feed back to asset prices. Crucially, we show this channel exhibits tail asymmetry -- it self-stabilizes in normal times but amplifies during stress.

Research Gap

- Rey et al. (2024) focus on equity-FX transmission but do not trace the effect through to hedging costs and bond flows.
- Nenova et al. (2025) study FX swap activity and yield curves but use OLS/linear methods that cannot capture tail amplification.
- No existing work applies quantile methods (quantile connectedness, Delta-CoVaR) to the equity-FX-hedging-bonds transmission chain.
- The offsetting forces hypothesis (yield curve vs FX effects on hedging costs) has not been formally tested across the quantile spectrum.

2. Research Questions

Primary Research Question

RQ1: Does the FX hedging activity of NBFIs constitute a contagion channel that transmits equity market shocks to cross-border bond flows? Specifically, does the transmission chain (equity shocks $\rightarrow$ FX movements $\rightarrow$ hedging cost changes $\rightarrow$ bond flow reversals $\rightarrow$ feedback) exhibit tail asymmetry -- self-stabilizing at the median but amplifying in the tails?

Secondary Questions

- RQ2: Do yield curve movements offset FX-driven hedging cost changes in normal times, and does this offset break down during stress? (Quantile regression test.)
- RQ3: What is the systemic risk contribution of the FX hedging channel? How much does the tail risk of cross-border bond flows increase when hedging conditions deteriorate? (Delta-CoVaR framework.)
- RQ4: Is the FX hedging transmission chain dormant in normal times but active during stress? (Quantile connectedness comparison: $\tau=0.05$ vs $\tau=0.50$.)
- RQ5: Does NBFi penetration in a country amplify its vulnerability to FX hedging cost shocks? (Panel regression with NBFi interaction, IV with MP shocks.)
- RQ6: Are spillovers bi-directional? Do European/Japanese NBFIs investing in US Treasuries on a hedged basis create reverse spillovers to the US?

3. The Economic Mechanism

3.1 Transmission Chain

The contagion operates through six sequential steps:

1. Shock hits equity markets (e.g., US rate hike, risk-off event, geopolitical shock).
2. Exchange rates move: The USD appreciates as foreign investors rebalance toward US equities (Rey, Stavrakeva & Tang 2024 -- 'currency centrality' mechanism). The USD plays a uniquely central role in this network.
3. FX hedging costs spike: CIP deviations widen as the FX swap basis blows out. NBFIs using short-term FX swaps to hedge long-term bond positions face rollover risk (Nenova, Schrimpf & Shin 2025).
4. NBFIs deleverage: Faced with higher hedging costs and potential margin calls on FX derivatives, NBFIs reduce hedged cross-border bond positions.
5. Bond sell-off: Cross-border bond outflows push up yields in the destination country, further widening the CIP basis and tightening financial conditions.
6. Feedback: Higher yields and wider spreads feed back to equity markets, restarting the cycle with geometric dampening.

3.2 The Offsetting Forces Hypothesis

A critical insight is that equity shocks and yield curve movements create opposing forces on FX hedging costs in normal times:

Force	Normal Times	Stress
Equity -> FX (Rey et al.)	USD appreciates moderately	USD appreciates violently (safe haven)
Yield curve (Nenova et al.)	Steepens -> attractive hedged carry	Flattens/inverts -> carry disappears
CIP deviations	Small -> manageable hedging cost	Blow out -> prohibitive hedging cost
Net effect	Partially offsetting -> self-stabilizing	All reinforcing -> amplification

Key prediction: The interaction term ($equity_return \times VIX$) in a quantile regression of CIP basis is insignificant at the median but large and significant in the tails. This is the formal test of tail amplification.

4. Data Requirements

Source	Variables	Use
Bloomberg / Refinitiv	Cross-currency basis (EUR, GBP, JPY, CHF, AUD vs USD)	CIP deviations / hedging costs
FRED / ECB / BoJ	Yield curve slopes (10Y - 2Y) by economy	Hedging demand determinants
MSCI / S&P	Equity index returns (US + foreign)	Equity shock identification
TIC (US Treasury)	Cross-border bond flows by investor type	Portfolio flow channel
BIS OTC Derivatives	FX swap outstanding by counterparty sector	NBFI hedging volume proxy
FSB NBFI Monitor	Sector-level assets by jurisdiction	NBFI penetration variable
CBOE	VIX	Global risk appetite
High-frequency MP	Gurkaynak et al. shocks, Kearns et al. (2023)	IV for yield curves / CIP

For the prototype implementation, we generate synthetic data that mirrors the structure, temporal patterns, and cross-sectional variation of these real sources. The synthetic DGP embeds the tail asymmetry mechanism to validate the econometric approach. All synthetic data uses fixed random seeds for reproducibility.

5. Empirical Framework

The framework has three integrated empirical layers, each targeting a different aspect of the FX hedging contagion channel, plus panel regressions and IV estimation.

5.1 Layer 1: Quantile Regression -- Offsetting Forces Test

Quantile regression of CIP basis on equity returns, yield slopes, and VIX:

$$Q_{\tau}(CIP_basis_t) = \alpha + \beta_1 * Equity_t + \beta_2 * SlopeDiff_t + \beta_3 * VIX_t + \beta_4 * (Equity_t \times VIX_t) + \epsilon_t$$

Estimated at $\tau = \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$.

Key test: β_4 (interaction) is small at the median but large in the tails.

- Formal asymmetry test: Wald test of $H_0: \beta_4(\tau=0.05) = \beta_4(\tau=0.50)$.
- Coefficient profile plots across the full quantile spectrum, with 95% confidence bands.
- Cross-currency comparison to test robustness (EUR, GBP, JPY, CHF, AUD vs USD).

5.2 Layer 2: Delta-CoVaR (Adrian & Brunnermeier 2016)

Estimate the systemic risk contribution of the FX hedging channel:

$$\begin{aligned} \Delta\text{-CoVaR} &= \text{CoVaR}(\text{bond_flows} \mid CIP_basis = \text{stressed}) \\ &\quad - \text{CoVaR}(\text{bond_flows} \mid CIP_basis = \text{median}) \end{aligned}$$

A large negative Delta-CoVaR means FX hedging stress significantly amplifies the tail risk of cross-border bond flows.

- Step 1: Estimate VaR of CIP basis using quantile regression on VIX and equity returns.
- Step 2: Estimate CoVaR of bond flows conditioned on CIP basis, VIX, and equity returns.
- Step 3: Compute Delta-CoVaR as difference between stressed and normal conditions.

5.3 Layer 3: Quantile Connectedness [Key Innovation]

Apply the Ando, Greenwood-Nimmo & Shin (2022) quantile connectedness framework to the 4-variable system:

$$\begin{aligned} \text{System: } &\{\text{US equity return, FX return, CIP basis, Bond flows}\} \\ \text{Estimated at } &\tau = \{0.05, 0.50, 0.95\} \end{aligned}$$

Generalized FEVD from QVAR yields quantile-specific connectedness.

Key prediction: Total connectedness at $\tau=0.05 \gg \tau=0.50$.

- FEVD heatmaps at each quantile showing directional spillovers.
- Cross-currency comparison of tail vs median connectedness.
- Rolling quantile connectedness to track time-varying tail spillovers.

5.4 NBF1 Penetration Panel Regressions

$$\begin{aligned} \text{bond_flows}_{\{c,t\}} &= \alpha_c + \beta_1 * CIP_basis_{\{c,t\}} \\ &\quad + \beta_2 * NBF1_{\{c,t\}} \end{aligned}$$

$$\begin{aligned}
 &+ \beta_3 * (CIP_{c,t} \times NBF_{c,t}) \\
 &+ \gamma * X_{c,t} + \epsilon_{c,t}
 \end{aligned}$$

If $\beta_3 < 0$: higher NBF penetration amplifies the negative effect of hedging cost shocks on cross-border bond flows.

5.5 IV Estimation

Instrument CIP basis / yield curve slopes with high-frequency monetary policy shocks (Gurkaynak et al. 2005, Kearns et al. 2023, Jarocinski 2024) and monetary policy-induced 'risk shifts' (Kroencke et al. 2021, Bauer et al. 2023). This follows Nenova et al. (2025) who use these instruments to identify causal effects of central bank announcements on FX hedging demand.

6. Expected Contributions

1. First complete model of the equity-FX-hedging-bonds contagion chain, combining Rey et al. (2024) and Nenova et al. (2025) into a unified empirical framework.
2. Discovery of the 'offsetting forces' mechanism: yield curve and FX effects on hedging costs partially cancel at the median but reinforce in the tails, creating nonlinear amplification detectable only with quantile methods.
3. First application of quantile connectedness (Ando et al. 2022) to the FX hedging transmission chain, showing that contagion activates asymmetrically during stress.
4. Evidence that the maturity mismatch in NBFi FX hedging (3-month swaps hedging 10-year bonds) creates systemic rollover risk visible in Delta-CoVaR measures.
5. Evidence on bi-directional spillovers: the FX hedging channel operates not only from the US outward but also in reverse, through European and Japanese NBFIs investing in US Treasuries on a hedged basis.
6. Policy-relevant finding: FX swap volumes should be monitored as a systemic risk indicator, and central bank FX swap lines can break the contagion cascade.

Positioning

This project directly extends the BIS research programme led by Hyun Song Shin. Nenova, Schrimpf & Shin (2025) establish the facts; we add the nonlinear econometric machinery (quantile methods) to detect tail amplification that their linear framework cannot capture. This positions the paper as a natural complement to BIS research, suitable for a BIS Working Paper or a finance journal.

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